

CS-2005 – Database Systems – Spring 2026

Assignment # 02

Maximum Marks: 10

Post Date: March 19, 2026

Due Date: April 6, 2026

Note:

- Plagiarism of any kind will not be tolerated.
- Late submissions will not be accepted.

Question 01

A Digital Bookstore maintains a database to manage information about customers, books, authors, and purchase transactions. Customers can purchase multiple books, and each book may be written by one or more authors. The bookstore records the quantity of each book purchased and the date of purchase. Management wants to analyze customer purchasing behavior and book pricing using SQL queries.

The database schema is given below:

Customers (customer_id, customer_name, city)

Books (book_id, title, price, publisher)

Authors (author_id, author_name)

Book_Authors (book_id, author_id)

Purchases (purchase_id, customer_id, purchase_date)

Purchase_Details (purchase_id, book_id, quantity)

Instruction:

Write SQL queries for the following requirements. Your solutions must make use of nested queries (including correlated conditions where appropriate) and aggregate functions.

A. Retrieve the names of customers who purchased books whose price is greater than the average price of all books in the system.

B. Display the titles of books whose price is greater than the average price of books published by the same publisher.

- C. List the names of customers who have purchased at least one book, but have never purchased any book published by 'Pearson'
- D. Display book titles that have never been purchased by any customer.
- E. Retrieve the names of authors who have written books priced higher than the average book price in the system.
- F. Display the customer_id and total quantity of books purchased by each customer, but only include those customers whose total purchased quantity is greater than 10.

Question 02

A City Transportation Authority maintains a database to store information about passengers, buses, routes, and ticket bookings. Each passenger can book multiple tickets, and each ticket is associated with a particular bus and route. The system records ticket fares and travel dates to analyze transportation usage patterns.

The database schema is given below:

Passengers (passenger_id, passenger_name, city)

Buses (bus_id, bus_name, capacity)

Routes (route_id, source_city, destination_city)

Tickets (ticket_id, passenger_id, bus_id, route_id, travel_date, fare)

Instruction:

Write SQL queries for the following tasks. All queries in this question must be solved using JOIN operations between tables.

- A. Display passenger names along with the bus name and route details for their bookings.
- B. Retrieve the passenger name, travel date, and fare for each ticket booked.
- C. Display the bus name and total number of passengers traveling on each bus.
- D. Show the source city, destination city, and passenger name for all booked tickets.
- E. Create a VIEW named PassengerTravelSummary that displays:
- passenger_name
 - bus_name
 - source_city
 - destination_city
 - travel_date

F . Write SQL statements to perform the following operations:

1. Add a column contact_number VARCHAR(15) in the Passengers table.
2. Modify the datatype of fare to DECIMAL(10,2).
3. Drop the column capacity from the Buses table.
4. Drop the table Routes from the database.

Question 03

A fitness app company wants to manage information about Workout Programs, Trainers, and Fitness Centers. The system should store program details, the trainers who run them, and the fitness centers hosting the programs.

System Description:

1. Workout Programs:
Each program has a unique ProgramID, ProgramName, Duration, and DifficultyLevel.
 2. Trainers:
Each trainer has a unique TrainerID, TrainerName, and ContactNumber.
 3. Trainer Assignment:
A program can be conducted by multiple trainers, and a trainer can run multiple programs. The Role of the trainer in a specific program (e.g., Lead, Assistant) must be recorded.
 4. Program Coordinator:
Each program has exactly one Coordinator, but coordinators are not stored as separate entities. The CoordinatorName is stored directly in the Program table.
 5. Fitness Centers:
Programs are offered at one or more Fitness Centers (CenterID, CenterName). A center can host multiple programs.
- a. Identify all primary entities and list their attributes. Clearly underline the Primary Key (PK) for each entity.
- b. Identify the main relationships between the entities and specify their cardinality (1:1, 1:M, M:N).
- c. Requirement 3 represents a many-to-many relationship. Identify the associative entity (linking table) required to resolve it and list all its attributes including the primary key.

d. Explain why ManagerName should not be modeled as a separate entity in this simplified design.

Question 04

Design an ER Schema for keeping track of information about Bechtree retail outlets across Pakistan, given the description below:

Bechtree operates retail outlets in 6 different provinces: Punjab – Bechtree Lahore, Sindh – Bechtree Karachi, Khyber Pakhtunkhwa – Bechtree Peshawar, Balochistan – Bechtree Quetta, Gilgit-Baltistan – Bechtree Gilgit, and Islamabad Capital Territory – Bechtree Islamabad. Each province has two major cities where outlets are located: Lahore and Faisalabad in Punjab, Karachi and Hyderabad in Sindh, Peshawar and Mardan in Khyber Pakhtunkhwa, Quetta and Gwadar in Balochistan, Gilgit and Skardu in Gilgit-Baltistan, and Islamabad and Rawalpindi in ICT.

Each city has a single main outlet office. You are required to design a centralized database to manage all Bechtree outlet operations across Pakistan.

Products have a unique ProductID, ProductName, Category, UnitPrice, and StockQuantity. Each product is available in one or more outlets.

Outlets have a unique OutletID, OutletName, City, Province, and ManagerName. Each outlet stocks multiple products.

Customers may come from any province or city in Pakistan. Each customer has a unique CustomerID, FirstName, LastName, MailingAddress, zero or more PhoneNumbers, and zero or more EmailAddresses. MailingAddress includes Street, City, Province, PostalCode. Email addresses are unique. Customers may purchase one or more products from different outlets.

Sales have a unique SaleID, SaleDate, OutletID, CustomerID, ProductID, QuantitySold, UnitPrice, TotalPrice, PaymentStatus (Paid, Partially Paid, Pending), and PaymentMethod.

Task: Design a conceptual ER diagram capturing all entities, their attributes, primary keys, and relationships. Clearly specify the cardinality of each relationship and indicate any associative entities required to resolve many-to-many relationships (e.g., products sold by multiple outlets).

Question 05

A company called TruthAI manages a system to detect fake news articles across multiple online platforms. The system tracks news articles, the AI models used to analyze them, journalists who submit articles, and fact-checking agencies that verify the content. Each article may be analyzed by multiple AI models, and each AI model can process multiple articles. Journalists may submit multiple articles, and each article may be verified by one or more agencies. For every verification, the system records the verification outcome, confidence score, and date of verification.

The system keeps the following information in a single unnormalized table:

- ArticleID, ArticleTitle, SubmissionDate
- JournalistID, JournalistName, JournalistEmail
- PlatformID, PlatformName
- AIModelID, AIModelName, ModelVersion
- AgencyID, AgencyName, AgencyPhone
- VerificationID, VerificationDate, VerificationOutcome, ConfidenceScore

Tasks:

1. Explain the insertion, update, and deletion anomalies that might occur if this table is kept as-is.
2. Identify the functional dependencies (FDs) among the attributes. State any assumptions you make.
3. Normalize the table to 3NF:
 - Show intermediate tables in 2NF and the dependencies removed.
 - Show final 3NF tables with primary keys (PK), foreign keys (FK), and alternate keys if any.
4. Highlight potential data redundancy problems, such as repeated agency phone numbers or AI model info, and the type of anomaly they may cause.

THE END!